Student Name :
Student ID:

<u>Choose the correct answer:</u>

Q1.

During the design of a new operating system, the development team first identifies the needs of users and the technical requirements of the system before selecting an implementation strategy. According to the lecture, this step is necessary because:

A. Operating system design depends only on the programming language.

B. Operating system design begins by defining goals and specifications.

C. The hardware determines all software requirements.

D. System calls must be implemented before defining system goals.

Q2.

A software engineer states that an operating system should be **easy to use, easy to learn, reliable, secure, and fast**. These characteristics represent:

A. Kernel objectives

B. Hardware requirements

C. User goals

D. System implementation strategies

Q3.

Which of the following characteristics is considered a **system goal** rather than a **user goal**?

A. Easy to learn

B. Reliable operation

C. Easy to maintain

D. Fast response time

Q4.

An operating system is implemented using multiple programming languages. Low-level hardware-dependent components are written in Assembly language, while most of the kernel is written in C.

What is the primary reason for using Assembly language?

A. It improves application performance.

B. It provides direct compatibility with hardware.

C. It simplifies memory management.

D. It replaces system calls.

Q5.

A modern operating system separates its functionality into multiple layers, where each layer interacts only with the layer directly beneath it.

What is the primary advantage of this design?

A. It eliminates the need for a kernel.

B. It improves modularity and simplifies maintenance.

C. It removes the need for system calls.

D. It maximizes execution speed by allowing unrestricted communication.

Q6.

Which operating system structure moves many operating system services from the kernel into user space to improve reliability, security, and portability?

A. Simple Structure (MS-DOS)

B. UNIX Structure

C. Layered Structure

D. Microkernel Structure

Q7.

In the traditional UNIX operating system, the kernel is responsible for:

A. Executing only user applications.

B. Providing services such as memory management, CPU scheduling, and file system management.

C. Replacing system programs.

D. Managing graphical user interfaces only.

Q8.

Which of the following best describes the architecture of **MS-DOS**?

A. It consists of independent layers with clear interfaces.

B. It has a highly modular microkernel architecture.

C. Its interfaces and functional levels are not well separated.

D. It places most services in user space.

Q9.

Which operating system structure may experience additional communication overhead because services execute in user space rather than within the kernel?

A. MS-DOS Structure

B. UNIX Structure

C. Layered Structure

D. Microkernel Structure

Q10.

A software architect recommends adopting a **Layered Operating System Structure** instead of a **Simple Structure**.

Which statement best justifies this recommendation?

A. Layered systems eliminate the need for hardware.

B. Layered systems organize the operating system into modules, making it easier to understand, debug, and maintain.

C. Layered systems require only Assembly language.

D. Layered systems do not use system calls.

Answer the following:

B. Draw a concept map illustrating the overall architecture of an Operating System.

C. Discuss the flow of system call between user mode and kernel mode, neat sketches are available